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Project Title:	Directed Assembly of Hydroxyapatite Scaffolds		
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Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Other		
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Zip Code:	61801	**Congressional District:**	15
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	PI Lewis and coPI Jamison have created 3-D periodic hydroxyapatite (HA) scaffolds with controlled multiscale porosity. Concentrated colloidal HA inks have been formulated with tailored viscoelastic properties required for direct-write assembly. 3-D scaffolds have been fabricated via robotic deposition of these inks in a layer-by-layer build sequence. PI Lewis has fully characterized with respect to their pore structure at the nanoscale (< 100 nm), microscale (100 nm to 100 microns), and mesoscale (> 100 microns) using both direct imaging techniques and indirect structural probes (e.g., mercury intrusion porosimety). co-PI Jamison has produced scaffolds of several designs in largequantities (> 100 scaffolds) to enable characterization and optimization of various structural attributes of interest. We focused specifically in year 2 on understanding the influence of interconnected microposity (~6 µm diameter) on initial mechanical properties of the scaffolds, degradation of the scaffolds in simulated biological conditions, and interaction of bone cells (osteoblasts) with these micropores in culture conditions. We established methods to reproducibly create this		

Task Description:	interconnected microporosity in HA scaffolds by loading the HA gel with fugitive polymethylmethacrylate (PMMA) beads during the directed deposition step of scaffold fabrication. We expect the knowledge and experience gained in producing diverse architectures of hydroxyapatite to provide valuable insights on producing similar scaffolds from polymers or polymer/HA composites. Such scaffolds could be used not only for bone repair, but for regeneration of skin and other tissues in the body. The versatility, simplicity, and compactness of the robotic deposition equipment used in this research would therefore meet NASA's long-term goal of creating a multifunctional materials platform that can be used as a controllable scaffold for the guided regeneration of a broad spectrum of tissue systems. The broad objectives of this ground-based research program are to fabricate and characterize mesoporous hydroxyapatite (HA) structures with microporous domains that serve as model tissue engineering scaffolds. Specifically, we will produce model HA scaffolds via a directed colloidal assembly technique known as robotic deposition. We will then characterize their cellular response through in vitro studies. Our ultimate goal is to develop a fundamental understanding of the interactions between bone tissue and HA scaffolds, and, from that understanding, establish guidelines for the rational design of hydroxyapatite scaffolds that optimize the rate and extent of tissue ingrowth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ability to produce engineered, 3-D scaffolds for bone generation could have a profound impact on surgical procedures involving bone grafts, which are currently highly invasive. Our research team has successfully fabricated 3-D periodic hydroxyapatite (HA) scaffolds via direct-write assembly of concentrated HA inks. Several scaffolds were created with varying lattice geometry, spacing, and connectivity in both 3-D periodic and 3-D quadrant architecture. By altering scaffold architecture, porogen content, and sintering conditions, 3-D HA scaffolds were produced with multiscale porosity to facilitate bone in-growth in both in vitro and in vivo studies. The largest pore channels (~ 150 – 750 microns), defined by the lattice geometry and spacing, are interconnected in all three dimensions within the scaffold. The intermediate pore channels (~ 5 microns), formed by porogen removal, are also interconnected in all three dimensions within the individual filamentary elements of the scaffold structure. Finally, the finest pore channels (< 1 micron), developed by incomplete sintering, can either be in the form of interconnected or closed porosity depending on the sintered density. Our research team also focused on the in vivo response to hydroxyapatite (HA) scaffolds and their efficacy for growth factor delivery in multiple animal models. We demonstrated rapid bone ingrowth in HA scaffolds that was nearly independent of their architecture. We also investigated their potential for load bearing bone scaffolds and in vitro drug elution. Results showed that the HA scaffolds improved bone regeneration compared to empty defects and that HA scaffolds with multiscale porosity were significantly more effective at delivering the osteoinductive growth factor BMP-2 than were scaffolds with macropores only.
	HA Scaffold Pore Structure: HA scaffolds with 3-D and quadrant architectures were created by direct writing of a concentrated hydroxyapatite ink. The largest pore channels (~ 150 – 750 microns), defined by the lattice geometry and spacing, are interconnected in all three dimensions within the scaffolds. The primary difference between the periodic and quadrant architecture, is that the latter structure enables three different channel spacings to be accommodated within the same scaffold. Intermediate pore channels (~ 5 microns), formed by porogen removal, are also interconnected in all three dimensions within the individual filamentary elements of the scaffold structure due to the high porogen loading within the inks used to print these scaffolds. Finally, the finest pore channels (< 1 micron), developed by incomplete sintering, can either be in the form of interconnected or closed porosity depending on the sintered density. Mercury intrusion porosimetry and surface profilometry techniques have been used to quantify the total pore volume, as well as relative pore volumes and characteristic pore sizes associated with each type of porosity (meso-, micro-, and nano-pores). Mechanical Properties of HA Scaffolds: We investigated the influence of interconnected microposity (~ 5 µm diameter) on initial mechanical properties of the scaffolds, degradation of the scaffolds in simulated biological conditions, and interaction of bone cells (osteoblasts) with these micropores in culture conditions. We established methods to reproducibly create this interconnected microporosity in HA scaffolds by loading the HA gel with fugitive polymethylmethacrylate (PMMA) beads during the directed deposition step of scaffold fabrication, which yield spherical pore replicas upon burnout. We designed and completed a series of in vitro experiments to simulate the degradation behavior of HA scaffolds having either dense or microporous structure. The compression strength and modulus of non-microporous scaffolds measured after 336 h of degradation were not statistically different from initial values. Initial and residual values of strength and stiffness for these scaffolds were 302 ± 55 Mpa and 5.1 ± 0.3 GPa respectively. The same properties for microporous scaffolds showed a monotonic decrease with degradation time from initial values of 10 ± 2 MPa and 0.5 ± 0.2 GPa to 6.0 ± 0.5 MPa and 0.3 ± 0.1 GPa. These values represent reductions of 40% over initial properties. The results of these experiments suggest that fully dense scaffolds exhibit strength and stiffness comparable to human cortical bone after simulated degradation, whereas fully microporous scaffolds do not. These results provide bounds for design of load-bearing scaffolds having a combination of dense and microporous elements. In vivo Bone Response of HA Scaffolds: The in vivo bone response of 3-D periodic hydroxyapatite (HA) scaffolds was investigated. Two groups of HA scaffolds (11-mm diameter x 3.5-mm thick) are fabricated by direct-write assembly of a concentrated HA ink. The scaffolds consist of cylindrical rods periodically arranged into four quadrants with varying separation distances between rods. In the first group, HA rods (250 µm in diameter) are patterned to create pore channels, whose areal dimensions are 250x250 µm in quadrant 1, 250x500 µm in quadrants 2 and 4, and 500x500 µm in quadrant 3. In the second group, HA rods (400 µm in diameter) are patterned to create pore channels, whose areal dimensions of 500x500 µm in quadrant 1, 500x750 µm in quadrants 2 and 4, and 750x750 µm in quadrant 3. Each group of scaffolds is partially densified by sintering at 1200°C prior to being implanted bilaterally in trephine defects of skeletally mature New Zealand White rabbits. Their tissue response is evaluated at 8 and 16 weeks using micro-computed tomography, histology, and scanning electron microscopy. New trabecular bone is conducted rapidly and efficiently across substantial distances within these patterned 3-D HA scaffolds. Our observations suggest that HA rods are first coated with a layer of new bone followed by subsequent scaffold infilling via outward and inward radial growth of the coated regions. Direct-write assembly of 3D periodic scaffolds composed of micro-porous HA rods arrayed to produce macro-pores that are size-matched to trabecular bone may represent an optimal strategy for bone repair and replacement structures.

Task Progress:	Cell-Scaffold Interactions: HA scaffolds fabricated with rods containing interconnected microporosity presented a significantly different surface morphology than identical scaffolds with rods comprised entirely of dense HA. We investigated the effects of these morphological differences on cellular interactions holding all other attributes of the scaffolds fixed. Scaffolds were cultured with mouse mesenchymal stem cells (D1 ORL UVA) in 10% bovine serum supplemented with ascorbic acid and ß-glycerolphosphate to induce osteoblastic differentiation. SEM examination of cells attached to the two types of scaffolds and PS showed direct involvement of cell processes with micropores. Cells attached to the microporous surfaces showed intimate and extensive engagement with microporosity. It was therefore surprising that measurements of cell number and alkaline phosphatase (ALP) expression (a phenotypic marker of osteoblasts activity) on these surfaces were lower than controls. ALP activity of cells was delayed and extracellular matrix expression was more substantial in these cases. This suggests that the surface morphology affects cells in unexpected ways. Growth factor delivery: One rationale for designing and characterizing HA scaffolds with microporous rods is the opportunity they present for delivery of growth factors to aid in the regeneration of tissue within and adjacent to the scaffold. The interconnected nature of this microporosity provides a relatively large volume and surface area to contain drugs. We have completed a series of in vitro experiments to characterize the uptake and release of a model drug in order to assess the feasibility of such a feature. We chose bovine serum albumin (BSA) as a model drug because of its relatively lower cost and similar molecular weight to bone morphogenetic protein 2 (BMP-2), a potent bone growth factor. Elution profiles of BSA from lyophilized and non-lyophilized microporous scaffolds displayed logarithmic behavior, suggesting a diffusion-dominated process as expected. At seven days the drug was completely released from both sample types with no statistically significant difference in release rates. These results suggest that lyophilization of drug-bearing scaffolds may be a possible route to cost-effective clinical application. The 7-day release period may be sufficient for therapeutic action of growth factors like BMP-2 which are know to be present in bone healing in the first 3-4 days. HA scaffolds for bone regeneration and growth factor delivery: We conducted an in vivo study in which cylindrical HA scaffolds were implanted into the metatarsal bone of goats. The first objective of the study was to determine if scaffolds with multiscale porosity, or with macropores between rods and micropores within the rods, would improve bone regeneration as compared to scaffolds with solid rods. We found no significant influence of microporosity on bone ingrowth, bone density, or degree of bone organization within the defect site. This seemed to contradict previously reported in vitro results with two dimensional HA samples, which indicated a significant difference between the two types of scaffolds in terms of alkaline phosphatase activity and extracellular matrix expression. The difference between the results is due to interactions that occur in three dimensional biological environments which cannot be accounted for during in vitro experiments. Although there was no discernable difference in healing between the microporous and nonmicroporous scaffolds, they each yielded significantly increased healing compared to empty defects. This is attributed to chemotaxis and cell proliferation due to an increased calcium concentration from partial dissolution of HA. The in vivo study described above was also used to determine if HA scaffolds with microporosity can be used for BMP-2 delivery to bone defects. BMP-2 is a potent osteoinductive protein that is commonly used to assist in bone regeneration in tissue engineering research. It is also approved by the FDA for some clinical applications. Microporous HA scaffolds doped with BMP-2 contained significantly more bone as compared to microporous scaffolds without BMP-2 at both 4 and 8 weeks. This result indicates that the interconnected micropores can effectively retain and release BMP-2 and therefore increase bone regeneration.

Bibliography Type:	Description: (Last Updated: 12/21/2007)
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